

HPV vaccination simulation model

Cost-benefit analysis of extending vaccination age cap within VHA

2025-02-21

Overview

- Choosing a simulation model
 - Agent-based model (ABM) vs Markov model (MM)
 - Advantages and disadvantages of choosing ABM
- Designing an ABM model
- Some initial results
- Next steps

Agent-based models

- Step 1: Simulate a heterogenous population of individuals (agents)
 - Agents.
- Step 2: Model actions and interactions with environment
 - Environment and Interactions.
- Step 3: Track trajectory over time.
 - Flexibility.

Agent-based models

- Step 1: Simulate a heterogenous population of individuals (agents)
 - Agents: men in VA + smoking habits + MSM + vaccine hesitancy.
- Step 2: Model actions and interactions with environment
 - Environment and Interactions: sexual contact network + vaccine hesitancy.
- Step 3: Track trajectory over time.
 - Flexibility: smoking habits change over time.

Designing an ABM

Attributes of agent

1. $20 \leq \text{Age} \leq 85$
2. Vaccination status: Y/N.
3. HPV status: None/Low risk/High risk.
4. Covariates: Vaccine hesitancy, Smoking history*, MSM behavior*.

Transition between:

1. Disease states: Healthy/Pre-cancer/Cancer/Dead.

* have not included in model for now.

Simulate $n = 100,000$ agents in 20 – 85 age range.
10 % are vaccinated.
No one is HPV+; all are healthy in the beginning.

Designing an ABM

Attributes of agent

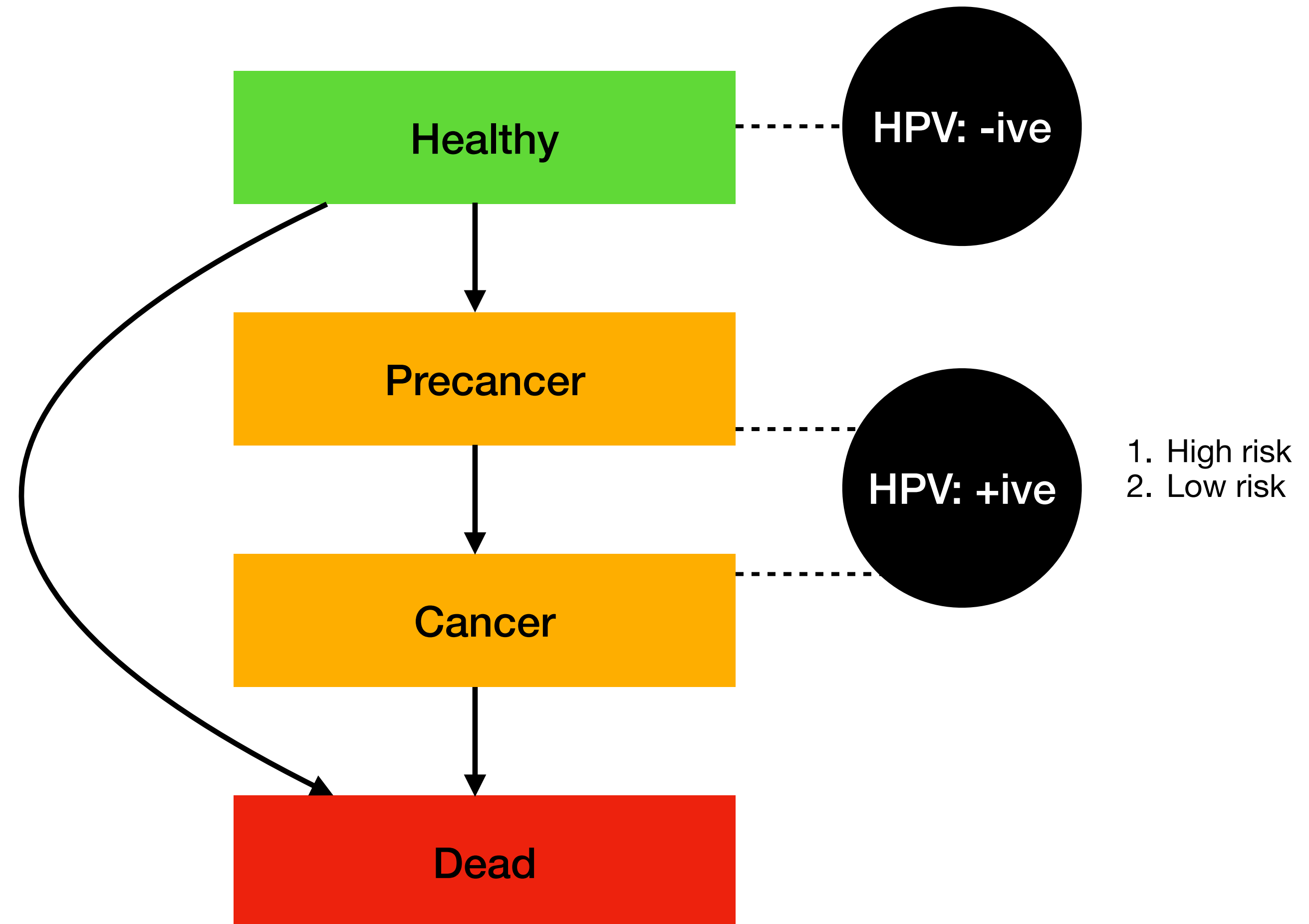
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Designing an ABM

Input: agent[age, vaccinated, hpv, disease], ageCap

Output: updated agent

1 age $\leftarrow$ age + 1;

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Start with agent with specified age, vaccination status, HPV status, and disease state.

Simulate $n = 100,000$ agents in 20 – 85 age range.
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No one is HPV+; all are healthy in the beginning.

Input: agent[age, vaccinated, hpv, disease], ageCap

Output: updated agent

```
1 age ← age + 1;  
2 if vaccinated = F then  
3   if age ≤ ageCap then  
4     P(vaccinated) ← 0.30  
5   if age > ageCap then  
6     P(vaccinated) ← 0.00  
7   Simulate vaccination
```

If agent does not have vaccine, check if eligible for vaccine
and simulate vaccination.

After simulation: vaccination = Yes/No

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4     P(vaccinated)  $\leftarrow$  0.30
5   if age > ageCap then
6     P(vaccinated)  $\leftarrow$  0.00
7   Simulate vaccination
8 if hpv = NONE then
9   P(infection)  $\leftarrow$  0.005 for vaccinated = T;
10  P(infection)  $\leftarrow$  0.05 for vaccinated = F.
11  Simulate infection
12  if infected = TRUE then
13    hpv = LOW with 0.80;
14    hpv = HIGH with 0.20.
15  else
16    Go to last line
```

If agent does not have HPV, simulate infection based on vaccination status. If not infected, go to next year.

If Infected, HPV status either low risk or high risk.

Designing an ABM

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11  Simulate infection
12  if infected = TRUE then
13    hpv = LOW with 0.80;
14    hpv = HIGH with 0.20.
15  else
16    Go to last line
17 if disease = HEALTHY then
18   if hpv = LOW then
19     P(disease = PRECANCER)  $\leftarrow$  0.05
20   else
21     P(disease = PRECANCER)  $\leftarrow$  0.07
22 else if disease = PRECANCER then
23   if hpv = LOW then
24     P(disease = CANCER)  $\leftarrow$  0.20
25   else
26     P(disease = CANCER)  $\leftarrow$  0.30
27 else
28   P(disease = DEAD)  $\leftarrow$  0.20
```

Based on HPV status, determine disease state transition probability.

Designing an ABM

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25   else
26     P(disease = CANCER) ← 0.30
27 else
28   P(disease = DEAD) ← 0.20
29 Simulate disease update (Healthy → Precancer → Cancer →
   Dead) with 1% background mortality + death at 100.
30 return agent;
```


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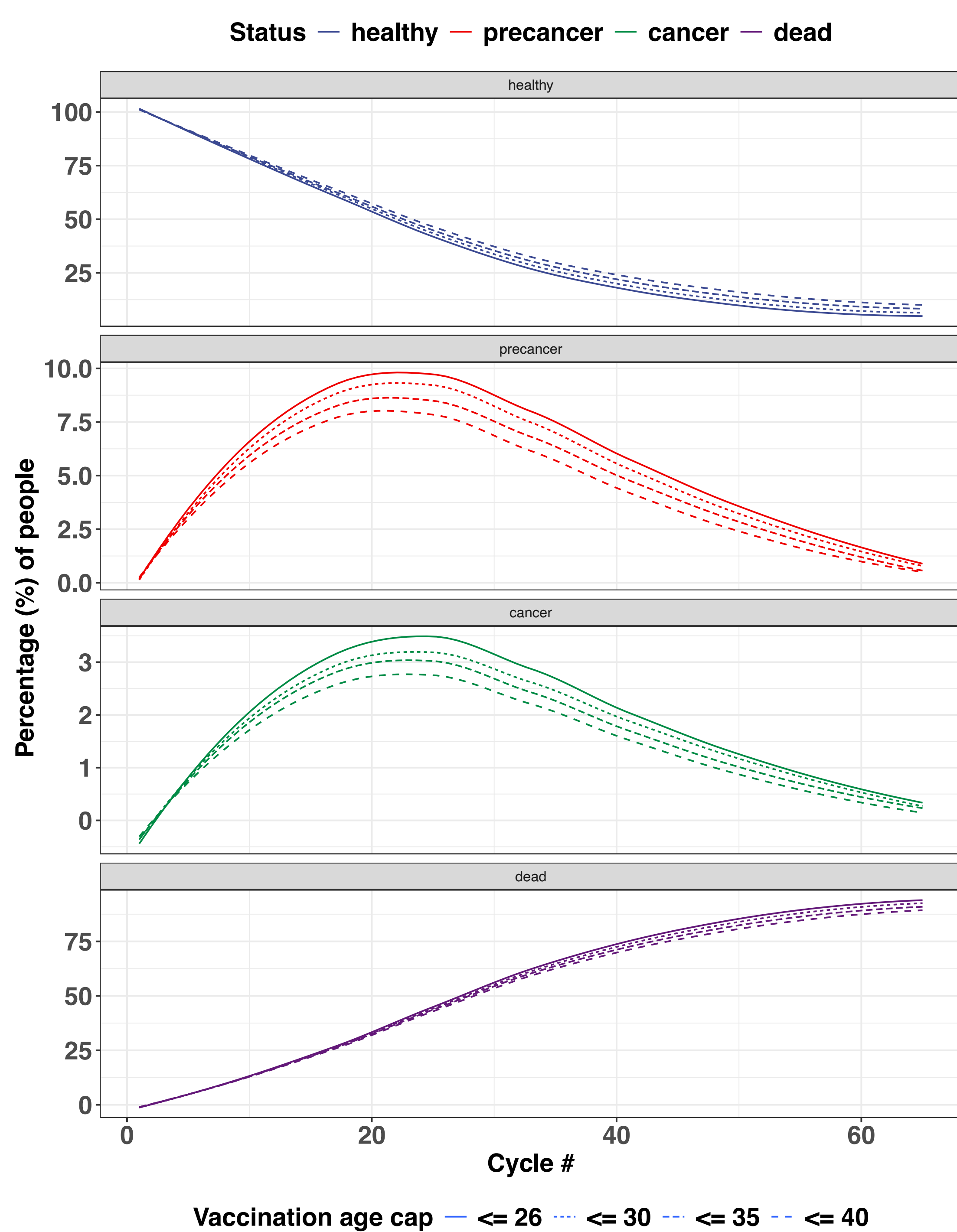
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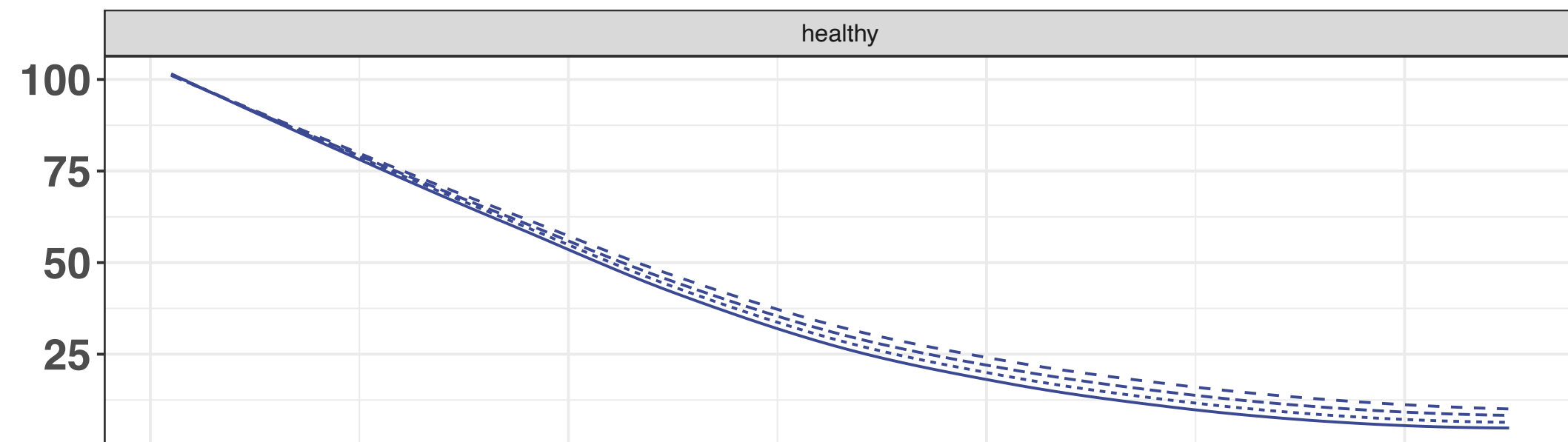
Run simulation for
ageCap = 26, 30, 35, 40



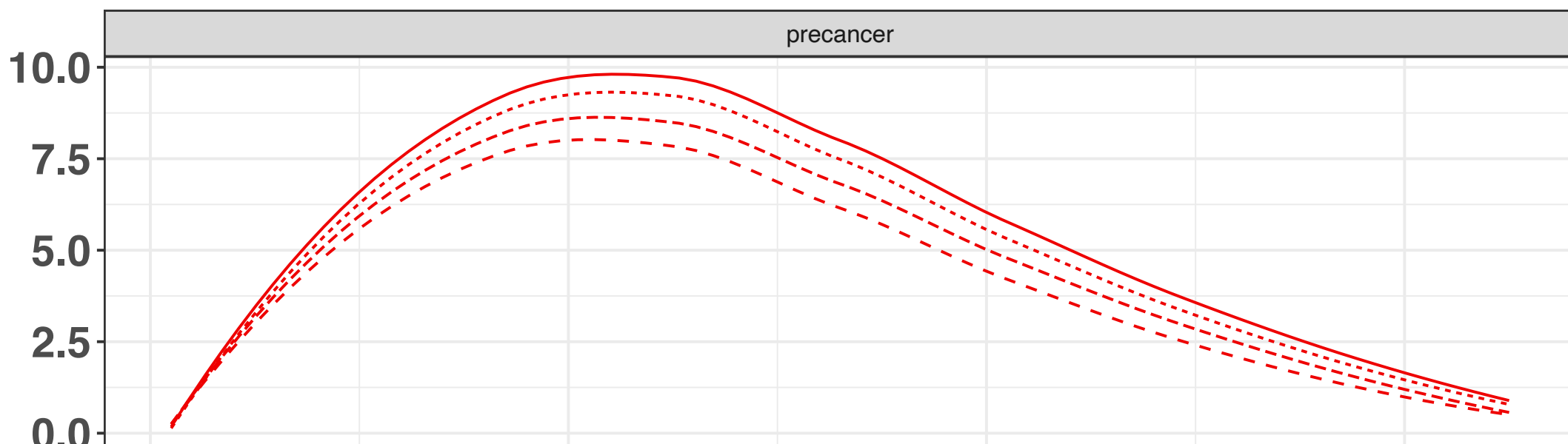
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ageCap = 26, 30, 35, 40

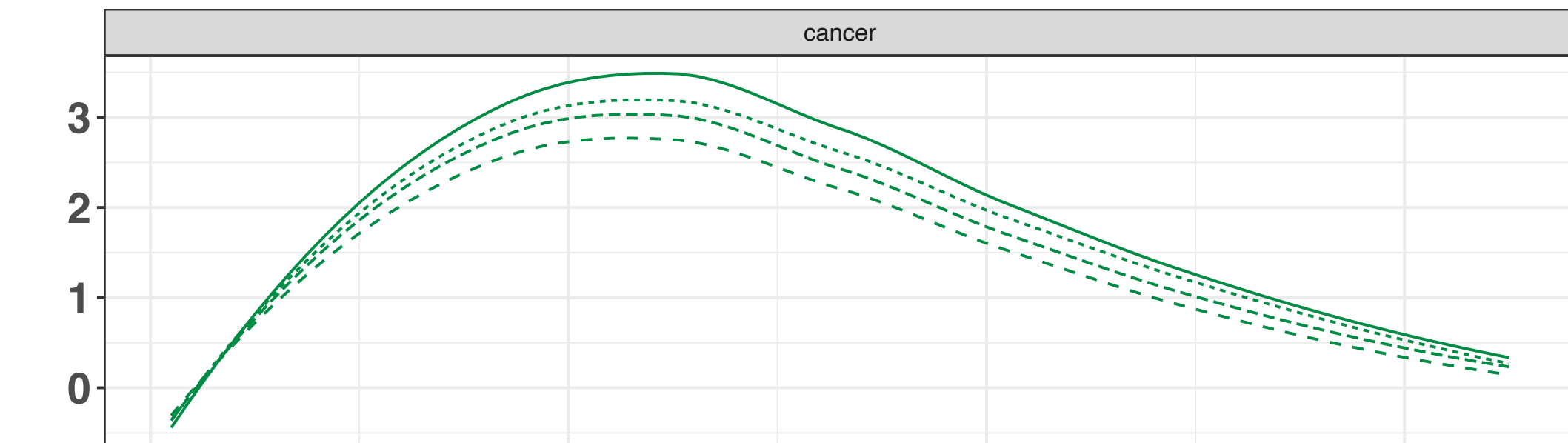
Status — healthy — precancer — cancer — dead



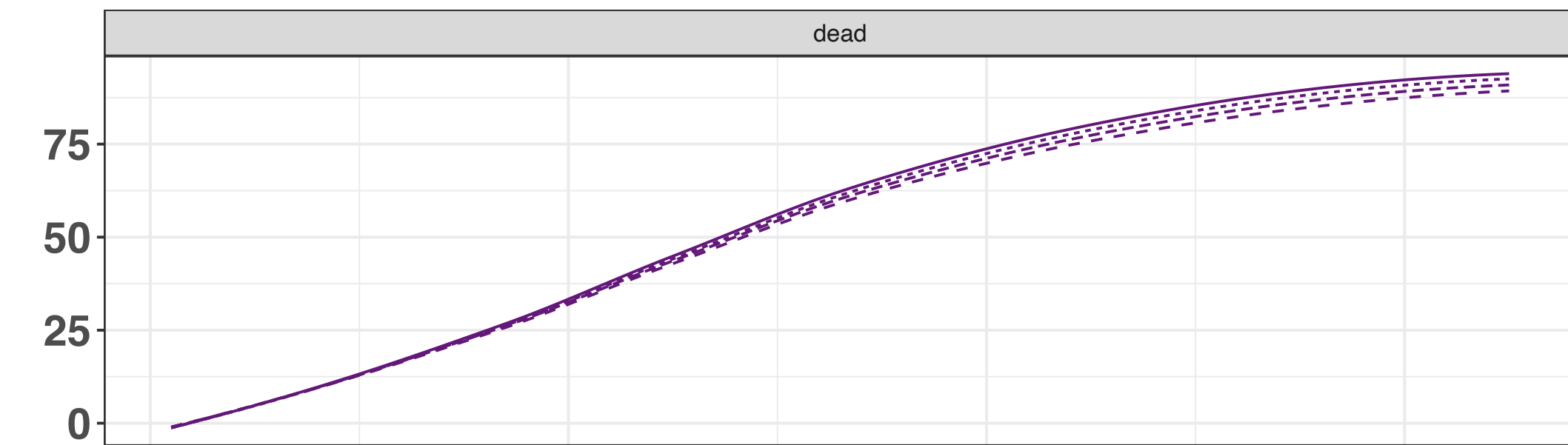
\$0/yr



\$100/yr

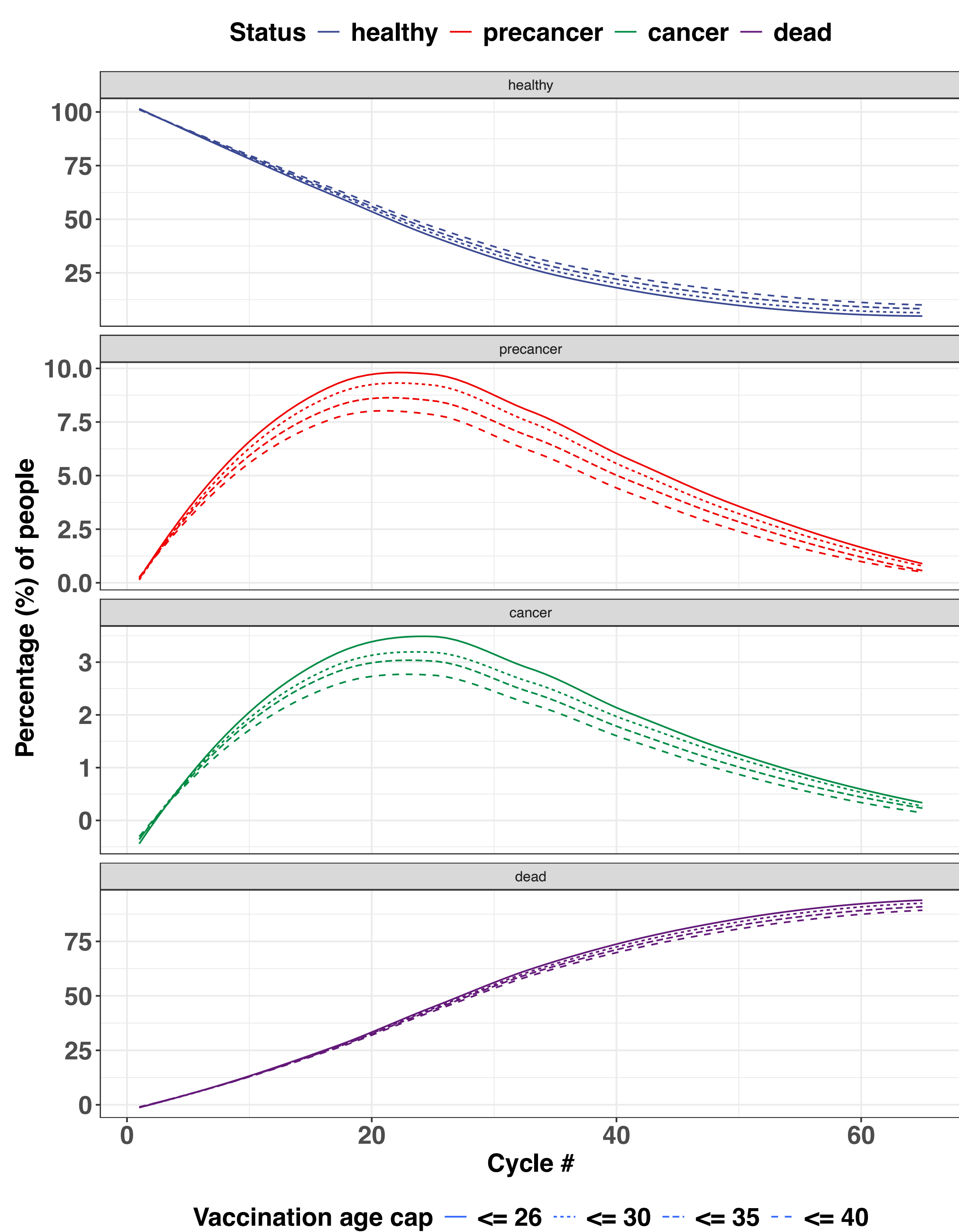


\$500/yr



\$0/yr

Vaccination age cap — <= 26 — <= 30 — <= 35 — <= 40

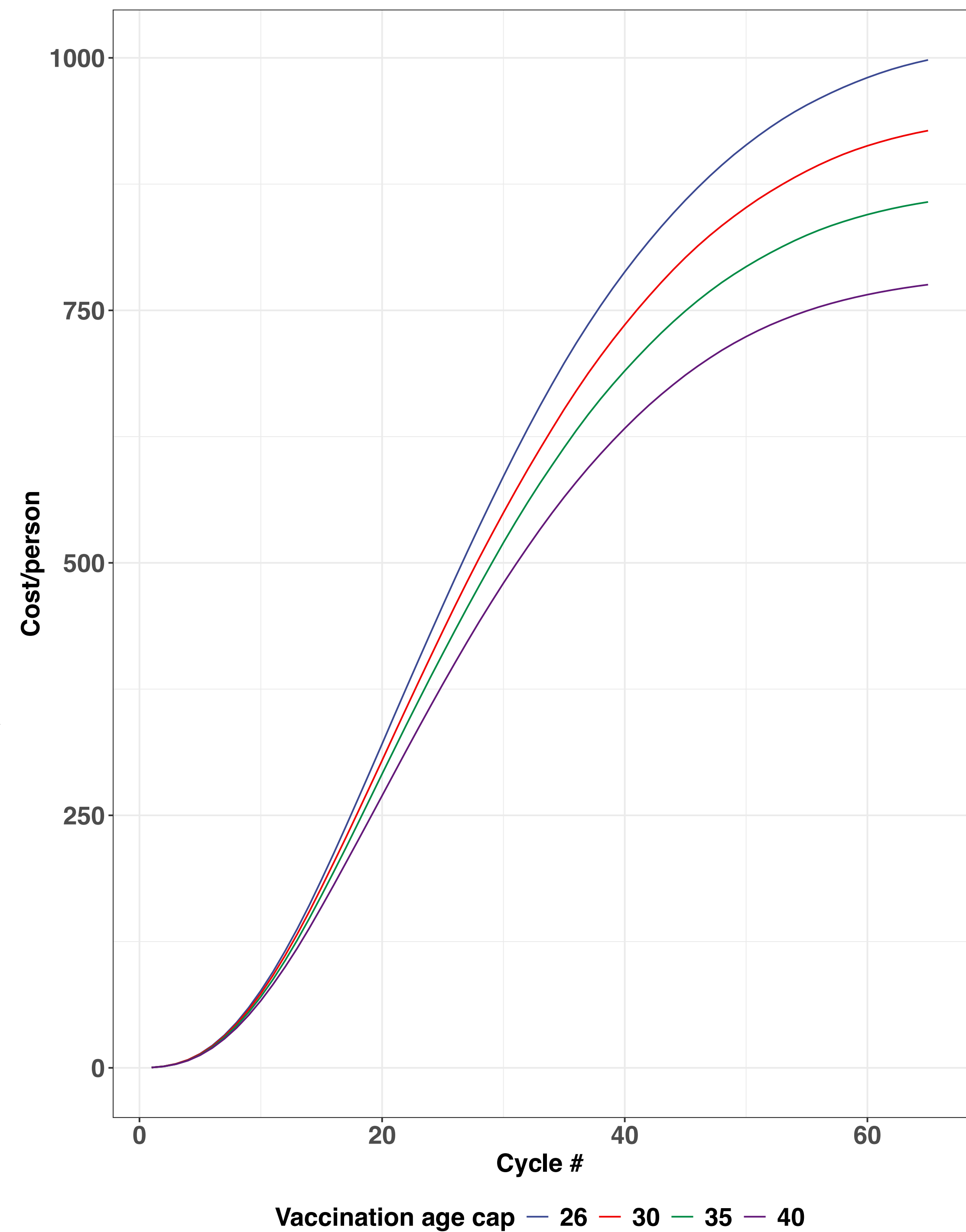


\$0/yr

\$100/yr

\$500/yr

\$0/yr



Next steps

To-do

- Add covariate information.
- Modify transition probabilities based on risk status.
- Calibrate parameters based on VA data.

Questions:

- Consider Markov model as well?
- Others?

ABM vs MM

Advantages

- Patient-level heterogeneity.
- Dynamic agent × environment interactions.
- Flexible over time.

- Simpler to design and implement.
- Computationally light.
- Data-light (summary data).

Disadvantages

- Computationally intensive.
- Data-intensive (patient-level data).

- “Memoryless” property not realistic.
- Limited heterogeneity.
- No agent × environment interactions.

ABM vs MM

Advantages

- Patient-level heterogeneity.
- Dynamic agent × environment interactions.
 - MSM behaviour.
- Flexible over time.
 - Smoking habits change over time.
 - Healthcare costs change over time.

- Simpler to design and implement.
- Computationally light.
- Data-light (summary data).

Disadvantages

- Computationally intensive.
 - Not too bad, simulation for 100k patients takes ~30 seconds on laptop.
- Data-intensive (patient-level data).
 - Access to VA data is an advantage, most studies don't have this.

- “Memoryless” property not realistic.
 - Prior smoking history important (?)
- Limited heterogeneity.
- No agent × environment interactions.

Designing an ABM

Attributes of agent

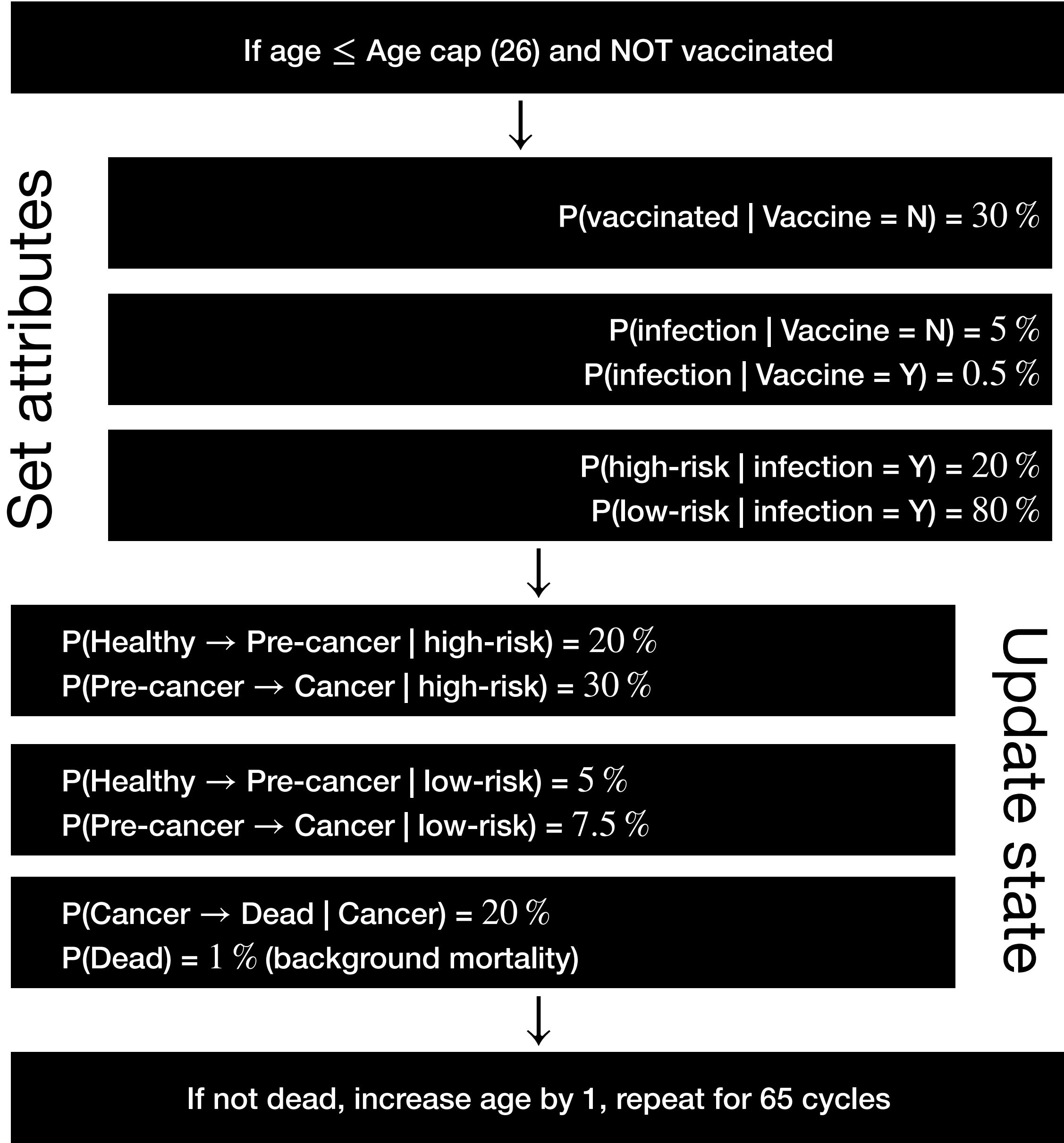
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Cost/year: healthy = 0, precancer = 100, cancer = 500